

A Controlled NISQ Ablation of Hybrid Quantum–Classical CNNs for Biomedical Image Classification

Guilherme H. Rodrigues^{1,*}, Ulisses Dias¹

¹School of Technology (FT), State University of Campinas (UNICAMP), Limeira–SP, Brazil

*Presenting author | g290200@dac.unicamp.br

Abstract

Does the quantum module of a hybrid quantum–classical CNN (H-QCNN) help biomedical image classification in the NISQ regime? Swapping *only* that module for a matched classical block (31,818 thermographic frames, 11 rats), we find **no aggregate advantage** and **significantly lower minority-class recall** for the quantum module (0.518 vs 0.777; $p = 0.008$).

Keywords: Quantum machine learning; hybrid quantum–classical models; variational quantum circuits; NISQ; controlled ablation; biomedical image classification; infrared thermography.

Introduction

Hybrid quantum–classical models probe possible advantages on NISQ hardware [1, 2], with parameterized circuits [3] as compact non-linear modules. Showing that the quantum module *itself* helps requires a matched classical comparator under an identical protocol. Matched ablations [4–6] mostly report advantages, but almost always with subject-dependent or unspecified splits; under subject-independent validation [7], the advantage proved sensitive to patient-level leakage.

Objectives

- Under a stricter protocol, swap *only* the module for a **differentiable classical block of equal role**.
- Validate **subject-independently** (Leave-One-Rat-Out, LORO) with paired tests, stressing a **rare class** ($\sim 4\%$) [8].
- Locate any effect with confusion and SHAP attribution analyses.

Methodology

Data and design. 31,818 thermographic frames from 11 Wistar rats over four exercise-intensity domains [8] (Fig. 1). Design choices (AngleEmbedding, BasicEntanglerLayers, C–Q–C topology) follow the dominant patterns of our PRISMA 2020 [9] review of 81 peer-reviewed primary studies.

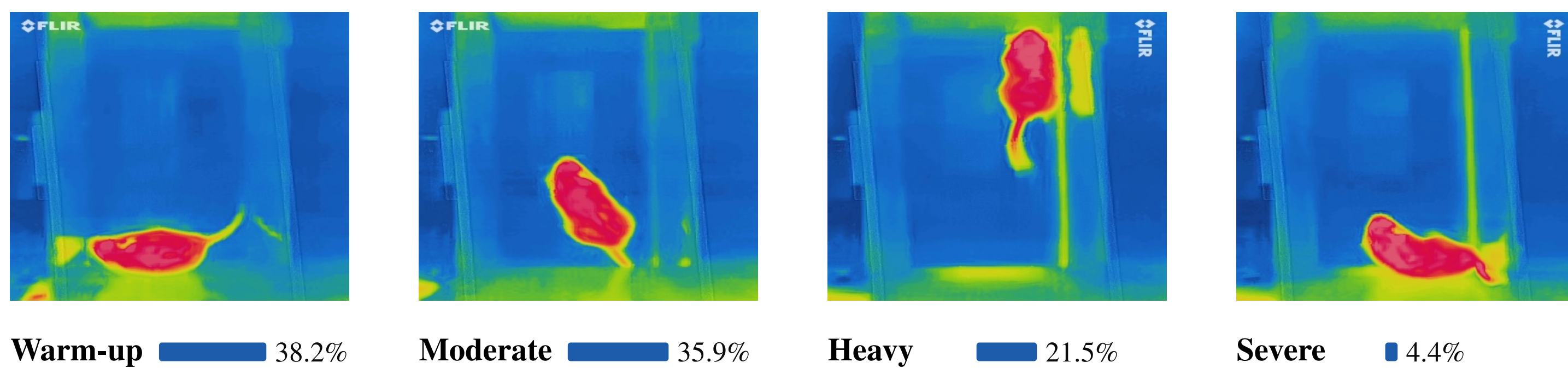


Figure 1: Representative frames per domain; bars show each domain's share of the 31,818 frames.

Architecture. Both models share the ResNet-50 [10] pipeline (Fig. 2); a non-differentiable comparator (e.g. a random forest) would block backbone fine-tuning.

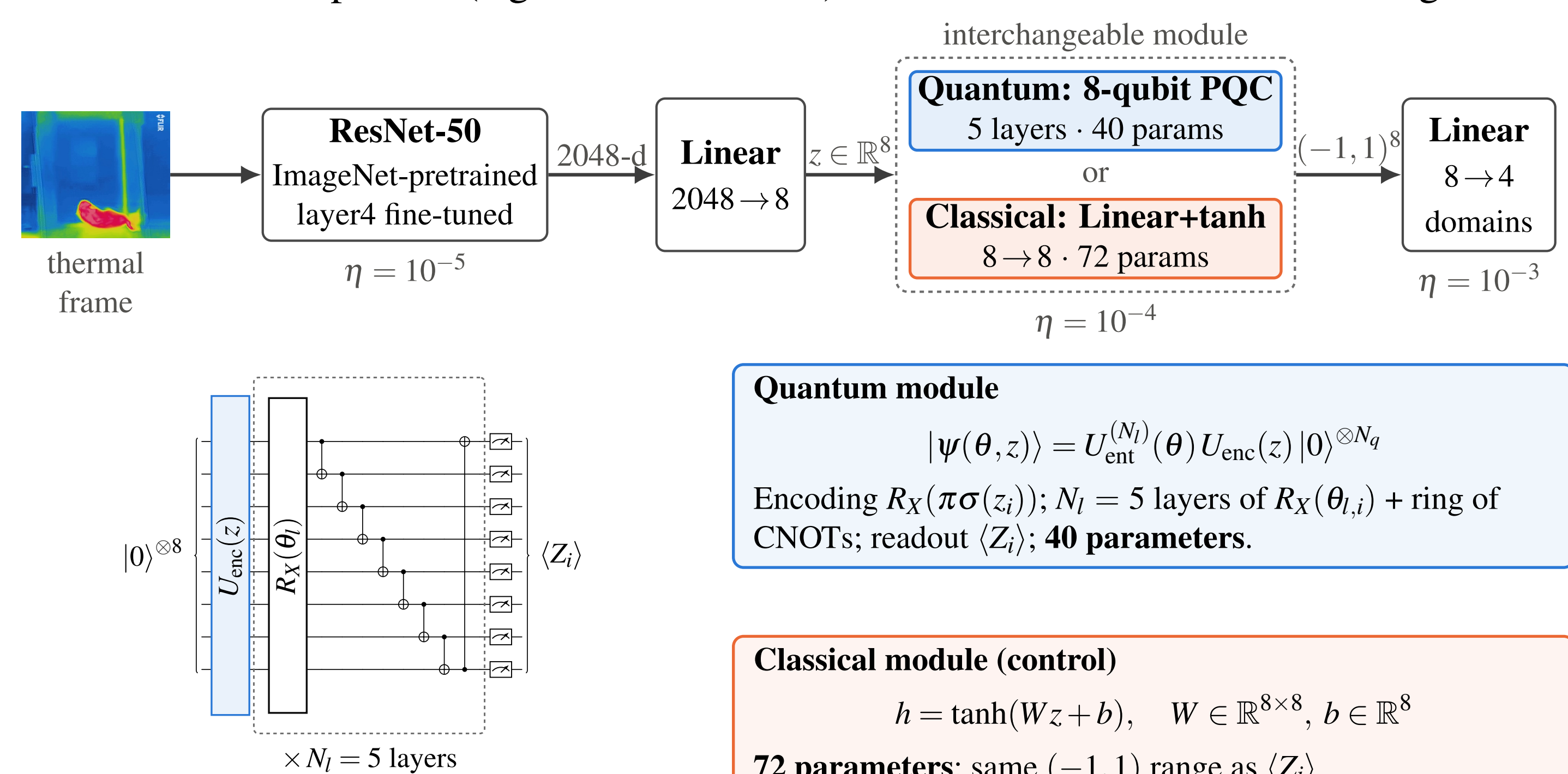


Figure 2: Paired ablation: shared pipeline (top); quantum circuit vs classical block (bottom).

Paired protocol. The two models are identical except for the module:

Training	class-weighted cross-entropy with label smoothing; Adam with per-role η (Fig. 2)
Held-out	the four severe-bearing rats (1, 9, 10, 11) $\times$ two seeds = 8 paired configurations (16 runs)
Validation	LORO (ten folds), test-time augmentation ($T = 5$), ensembling
Statistics	paired Wilcoxon signed-rank test, over configurations and over the 80 folds
Backend	PennyLane (default.qubit) + PyTorch, state-vector simulation

Results

Table 1: Mean over four held-out rats $\times$ two seeds ($n = 8$); paired Wilcoxon p , quantum vs classical.

Model	Bal. Acc.	F1	Prec.	Rec.	Sev. Rec.
Quantum (PQC, 40p)	0.529	0.438	0.541	0.529	0.518
Classical (matched, 72p)	0.532	0.411	0.492	0.532	0.777
Paired p (Wilcoxon)	0.74	0.38	—	0.74	0.008

Overall metrics. The two modules score the same. The quantum module wins balanced accuracy in only 2 of 8 configurations, its macro-F1 edge is not significant (fold-level $p = 0.20$), and the large standard deviations come from differences between rats.

Minority class. The only significant difference **favours the classical module**, better in **all eight** configurations (severe recall 0.777 vs 0.518; $p = 0.008$) — a shift in operating point, not in separability (Fig. 3).

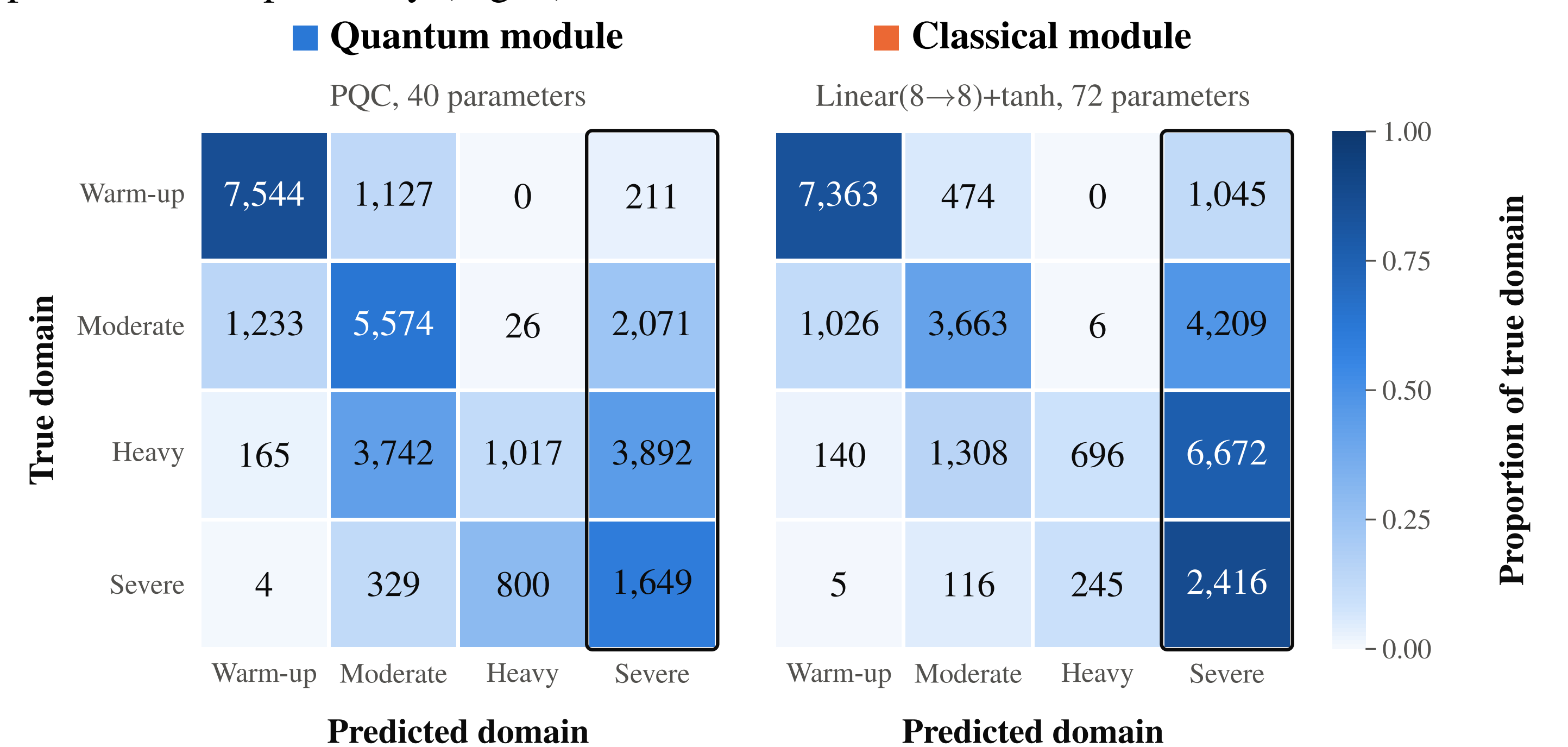


Figure 3: Pooled confusion matrices (counts; colour: row-normalized). Both modules miss *heavy* (recall < 0.12); the classical module predicts *severe* more often (outlined; precision 0.17 vs 0.21).

Subject dependence and attribution. The quantum module's only balanced-accuracy win is on rat 1 (the former fixed test subject); it reverses on the other three, one with zero severe recall. The matched classical block reproduces, and beats, the minority-class recovery once credited to the hybrid pipeline; contrastive SHAP maps [11] place the effect in the **fine-tuned backbone**, not in the quantum circuit.

Conclusions

In this NISQ regime, the 40-parameter PQC confers **no measurable advantage** over a matched differentiable classical block, and is **significantly worse on the minority class**.

- Apparent gains stem from fine-tuning the classical backbone; with [7], this indicates that reported H-QCNN advantages are contingent on the validation protocol.
- Contribution: subject-independent, imbalance-aware evidence and an attribution that localizes the effect classically.
- **Next:** physical NISQ hardware via Amazon Braket (superconducting and trapped-ion QPUs) with zero-noise extrapolation.

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